Technology Stack

Date	15 March 2026
Team ID	xxxxxx
Project Name	EcoTrack – AI-Powered Carbon Footprint Tracker
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	**React.js, HTML5, CSS3, JavaScript, Tailwind CSS, Chart.js**	Provides a responsive and interactive user interface with reusable components, data visualization, and a modern user experience.
2	Backend / Server-Side	**Flask (Python)**	Handles API requests, carbon emission calculations, AI integration, and communication between the frontend and cloud services.
3	Database / Data Storage	**Firebase Firestore & Firebase Authentication**	Stores user profiles, daily habit logs, carbon history, and badges securely with real-time cloud synchronization.

4	Cloud / Hosting / Deployment	**Firebase, Vite Development Server, Flask Local Server**	Enables secure cloud database access, authentication, and efficient development and deployment of the application.
5	Version Control & CI/CD	**Git & GitHub**	Supports version control, team collaboration, source code management, and project tracking.
6	Third-Party APIs / Other Tools	**Groq API (LLaMA 3.3-70B Versatile), Carbon Interface API, Axios**	Groq API generates personalized eco-friendly recommendations, Carbon Interface API provides real-world emission factors, and Axios enables communication between frontend and backend services.